

An exact support bound for uniform triple minima

Research proof; independently reviewed

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Let a probability distribution on permutations of $[n]$ have q support points, all of positive weight. Require only that each member of every three-element subset is its earliest element with probability $1/3$. Pairwise rank balance is not assumed.

Theorem 1. *For $n \geq 4$, one has $q \geq n - 1$. Arbitrary positive weights and repeated permutations are allowed (repetitions can be merged). The bound is attained at $n = 4$ by the three ordered lists 1203, 3201, and 0213, with equal weights.*

Proof. Fix a pivot i , and for $j \neq i$ let v_j be the binary indicator of i preceding j , viewed in the q -dimensional weighted Euclidean space of support points. Write $p_{ij} = \Pr(i < j)$. The Gram matrix on the $m = n - 1$ columns is

$$G_i = D_i + \frac{1}{3}\mathbf{1}\mathbf{1}^T, \quad (D_i)_{jj} = p_{ij} - \frac{1}{3} \geq 0.$$

The off-diagonal entries follow from triple-minimum balance. The inequality on the diagonal follows because being earliest in a triple containing i, j implies $i < j$.

Suppose $q \leq n - 2$, so $m > q$. If every diagonal entry of D_i were positive, then G_i would be positive definite of rank m , impossible in a q -dimensional space. Thus some $a \neq i$ has $p_{ia} = 1/3$. For every $j \notin \{i, a\}$,

$$\Pr(i < a, i > j) = \Pr(i < a) - \Pr(i < a, i < j) = 0.$$

Consequently the event $i < a$ implies that i is globally first, up to a null event. In particular $\Pr(i \text{ globally first}) \geq 1/3$. This holds for every pivot i . Summing contradicts $n \geq 4$. The displayed example is checked by restricting to its four triples. $\square$

Attribution and scope. The condition is called weakly three-restricted min-wise independence. The argument uses the classical diagonal-plus-constant Gram method. A 2011 abstract by M. Vsemirnov states the more general bound $q \geq (n - k + 2)/(k - 1)$ for weakly k -restricted families; see the primary conference booklet, p. 40. Priority of the sharper special-case application above has not been established. It is a bound for permutation distributions, not for general dice. For the concatenated two-face palindrome construction it implies at least $2n - 2$ faces per die whenever triple fairness is exact.